

Req ID	Description
5	Hook to detect hiding_stat_init() behavior: monitor stat-related syscalls for rootkit detection
6	Hook to detect hooking_audit_init() behavior: monitor audit subsystem modifications
7	Python script parses process events and writes structured JSON log files
8	Hook file-open events on sensitive paths (/etc/passwd, /etc/shadow, SSH keys) in the LKM
9	Hook to detect taskstats_hook_init() behavior: monitor task statistics interface tampering

10	Create Makefile and automation scripts to compile, load, and run all hooks systematically
TOTAL	

Sub-Requirements	Story Points
5.1 Hook stat-family syscalls (stat, lstat, fstat, newfstatat) using kprobes 5.2 Capture syscall arguments: pathname, file descriptor, return values 5.3 Detect file/process hiding patterns (ENOENT for existing files) 5.4 Monitor /proc stat calls for inconsistent results 5.5 Track stat failure frequency per process for anomaly detection 5.6 Log events with timestamp, PID, process name, target path, severity 5.7 Export events through kernel-to-user communication channel	8
6.1 Hook audit-related kernel functions (audit_log, audit_log_start) 6.2 Monitor audit configuration changes via auditctl or direct manipulation 6.3 Track audit rule additions, deletions, and modifications 6.4 Detect audit daemon (auditd) termination or interference 6.5 Log all audit subsystem interactions with process credentials (UID, EUID) 6.6 Generate alerts for unauthorized audit modifications	5
7.1 Read raw events from kernel communication channel (/proc/lkvm) 7.2 Parse binary/text event format into Python data structures 7.3 Handle malformed or incomplete events gracefully 7.4 Define JSON schema: event_type, timestamp, pid, ppid, uid, comm, details 7.5 Write JSON logs to configurable output directory 7.6 Implement log rotation by size or time interval 7.7 Ensure atomic writes and buffer events for burst handling	3
8.1 Hook do_sys_open, do_sys_openat, or filp_open kernel functions 8.2 Capture file path, open flags (read/write/append), and mode 8.3 Resolve relative paths to absolute paths for matching 8.4 Monitor /etc/passwd, /etc/shadow, ~/.ssh/, /etc/ssh/, /etc/sudoers 8.5 Support configurable sensitive path list 8.6 Log accessing process PID, UID, GID, executable path, parent info 8.7 Flag write attempts and access from unexpected processes 8.8 Capture timestamps with nanosecond precision	8
9.1 Hook taskstats_exit and related process accounting functions 9.2 Monitor taskstats netlink interface for modifications 9.3 Detect attempts to hide process resource usage 9.4 Monitor for inconsistencies between taskstats and /proc data 9.5 Track unusual patterns in task accounting data 9.6 Export events through common communication channel 9.7 Follow established event format for consistency with other hooks	5

10.1 Create top-level Makefile to build all kernel modules 10.2 Support individual module targets (make hook_stat, make hook_audit, etc.) 10.3 Include make clean and make all targets 10.4 Create load_all.sh to load all hooks in correct order 10.5 Create unload_all.sh for safe hook removal 10.6 Create status.sh to check loaded module status 10.7 Create run_monitor.sh to start modules and Python parser together 10.8 Add root privilege checks and error handling to all scripts	5
	34

Priority	Sprint No.
5	2
6	2
7	2
8	2
9	2

10	2